Requirement Document: N8N Hands-On Beginner Course

Topic 3: Working with Core Nodes & Data Mapping

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 3 - Working with Core Nodes & Data Mapping
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
Last Updated	June 20, 2026

2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees/employees how to use the Edit Fields (Set) node to define new fields, modify existing values, assign specific data types (string, number, boolean), and map values dynamically from preceding nodes using expressions.

2.2 Learning Scope

In Scope:

- Adding fields of various types (String, Number, Boolean) to a workflow payload.
- Editing/overwriting values of pre-existing fields within the data stream.
- Using the n8n expression editor (`{{ ... }}`) to reference parent node output variables.
- Mapping a value from an input field to create a new output field.

Out of Scope:

- Complex JavaScript operations or loops within the expression block.
- Manipulating nested JSON arrays or objects.
- Data formatting functions (dates, timezone conversion, string splits).

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 and Topic 2.
- Understanding of basic JSON structure (keys and values).
- Understanding of standard data types (String, Number, Boolean).

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Create Initial Data	The workflow shall start with a node configured to output three distinct data fields representing a product: a string, a number, and a boolean.	**High**
REQ-002	Data Types	The initial fields shall be strictly typed inside the first node: • `productName` (String) • `basePrice` (Number) • `isAvailable` (Boolean).	**High**
REQ-003	Modify Data	A second connected node shall modify the value of the incoming `isAvailable` boolean field from `true` to `false`.	**High**
REQ-004	Variable Mapping	The second node shall create a new field named `inventoryLabel` and populate it by dynamically mapping the value of the incoming `productName` field.	**High**
REQ-005	Calculated Field	The second node shall create a new field named `priceWithTax` that calculates 10% tax on top of the incoming `basePrice` (using an expression: `basePrice * 1.1`).	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-006	Schema Verification	The final workflow execution shall output a single JSON object containing: `productName`, `basePrice`, `isAvailable`, `inventoryLabel`, and `priceWithTax`.	**High**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Expression Engine	The n8n expression parser must resolve variables in real time using the format: `{{ $json.fieldName }}`.
TR-002	Node Selection	The workflow shall use the Edit Fields (Set) node or the Set node depending on the n8n version, but using the same basic parameter schema.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn how data typing and dynamic referencing works so they can construct functional business workflows.
Instructor / Reviewer	Evaluation	Needs to confirm the trainee hasn't hardcoded mapped values and has instead used dynamic expressions correctly.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** Trainees understand that expression calculations (e.g. math operations) must be wrapped in `{{ }}` curly braces to evaluate in n8n.
- **Constraint 1 (CON-001):** The calculations in this beginner exercise are basic mathematics and do not account for floating-point rounding corrections.